

What are ethics?

The rules of right and wrong behavior. *Merriam-Webster* defines ethics as “the principles of way governing an individual or group.”

The Three Categories of Ethics

- ▶ Professional– usually defined by your employer of company; these are general guidelines
- ▶ Social– defined by society
- ▶ Individual– defined by personal heritage (inheritance) and integral (primary/principle) family values

Information Ethics

- ▶ Rules that define right and wrong behavior in the computing professions.
- ▶ The basis of trust and cooperation among workers and organizations
- ▶ In most cases, these are defined by the “owner” of the computer.

Ethics vs. Laws

- ▶ Ethics often precede laws.
- ▶ Laws exist to protect software developers (copyright and licensing) and users (privacy) and there are punishments for breaking the law.
- ▶ There are no “global punishments” for ethics violations.

Copyrights

The right of an author to copy, resell, lease, lend (provide), rent and or distribute material. Copyrights did not exist when software was first being developed.

Federal Digital Millennium Copyright Act (DMCA)

- ▶ The copyright law that is currently in effect (outcome).
- ▶ Any software or data created with a computer is owned (maintained) by its creator and is automatically copyrighted as soon as it has been recorded and is in “tangible” form.
- ▶ Without specific permission, it is illegal to copy software.
- ▶ Software designed by an employee is owned by the company.

Software Licensing

- ▶ License agreements are contracts between the purchaser and copyright owner.
- ▶ Many software companies offer site licenses or special licenses for companies, schools or government institutions.

Four Types of Software Licenses

- ▶ Public domain
- ▶ Freeware
- ▶ Shareware
- ▶ “All rights reserved”

Public Domain

Public domain software is not protected by copyright law; it is software that was either created with public funds (and is therefore already owned by the public) or the creator has forfeited (pay for) these rights to the public. Public domain software is pretty rare and is often unreliable (i.e., poor quality, virus-laden) since nobody's keeping track of it.

Freeware

Freeware is copyrighted software that is licensed to be copied and distributed without charge. Some freeware (Pegasus Mail, Eudora Light, Netscape, Internet Explorer, etc.) is really good stuff, but it's not in the public domain. That is, freeware is free only because its license says it is; the software is still under the owner's control. Thus, most freeware licenses prohibit the sale or modification of the software without the owner's permission, and they often limit the groups that are able to get the software without paying.

Shareware

Shareware is the same as freeware, except that the owner specifies a purchase fee for permanent use. That is, the software is licensed for copying and sharing without charge, but only for evaluation purposes. Anyone who decides to use the software long-term is to pay a specified fee to the owner. There is usually no official system for collecting these fees; users are expected to pay on their own (honor system) in exchange for documentation, a more powerful version of the software, or some other enticement. More and more shareware is programmed to turn itself off after a certain period of time (30 days is typical) unless the owner buys a special activation code from the copyright owner. Most of the “free” software in the world is actually shareware.

Software license with all rights reserved

Licensed only for use by purchasers. Almost all the really good stuff (Word, Oracle, Java, PhotoShop, Windows etc.) is licensed this way. You can't legally use it or even possess it without the owner's permission, usually defined very carefully in the license agreement.

Typical License Agreement

- ▶ You can install and operate the software on only one PC at a time.
- ▶ You can make only one back-up copy of the software.
- ▶ If you give the software to anyone, you also give up permission to use the software.
- ▶ You are not allowed to modify the software in any way.

Software Piracy

- ▶ The illegal copying of software
- ▶ The estimated loss due to piracy is approximately \$39 billion per year globally.
- ▶ Conviction on the first offense can result in a fine of up to \$250,000 and a jail sentence of up to five years. (Bowyer, *Ethics & Computing*, 1996)

Ethically and legally, society should recognize that individuals creating “intellectual property” have the right to receive credit and be protected from the incorrect use of their creations.

Intellectual Property

The result of intellectual activities in the arts, science and industry (Beekman, *Computer Confluence*, Prentice Hall, 2001) such is the property of the creator.

Fair Use

- ▶ provisions that let you paraphrase or quote small portions of someone else's work, provided that you give proper credit to the author for the material you use
- ▶ If you don't give credit, you will probably give at least some of your readers the impression that the material you borrowed originated with you. That's bad. It's so bad, it has a name, plagiarism, and it's a form of fraud.

Giving Credit

- ▶ Add an in-text reference to the author and publication year of the work used right after the borrowed material in the paper.
- ▶ Add a full citation of the work (author, title, publisher, publication year, etc.) at the end of your paper.

Electronic Information

- ▶ More and more information is available via CD-ROMs, the Worldwide Web, online databases, and other electronic communication channels
- ▶ Quoting and citing information is just as important with computer sources as with traditional written works. Even though computers and the Internet allow easy access for cutting and pasting items, it is considered plagiarism when the creator has not given permission and/or been given credit for his/her work.

Misuse of Information

- ▶ When using intellectual property, it should be used in the context originally intended and presented correctly so the meaning is not changed from the original intent.
- ▶ Blatant misuse of information in which the user knows he/she is plagiarizing but chooses to use the information anyway is an example of fraud. The *Merriam-Webster's Collegiate Dictionary* defines fraud as "the act of deceiving or misrepresenting" (Merriam-Webster, Inc. <http://www.m-w.com/cgi-bin/dictionary>, "fraud", 2001). Accidental plagiarism is a form of negligence. Negligence occurs when an individual is unaware or ignorant that he/she has misused intellectual property. Both fraud and negligence are ethically wrong and illegal.

Privacy

Information can be categorized as that which is public and that which is private. Public information is assumed to be able to be used without permission. Private information cannot be legally used without permission from its owner.

Inappropriate Use of E-mail

- ▶ Spamming
- ▶ Flaming
- ▶ Trolling

Spamming

Repeated posting or mailing of significantly identical messages. The emphasis is on the multiple sending, either many copies to one destination, or one copy to many destinations.

Flaming

Email message that expresses a strong opinion or criticism. Flames can be fun when they allow people to open their feelings, then return to the topic at hand. Others are simply insulting and can lead to flame (fire/spirit) wars. To engage in an online argument usually involving unsupported/groundless personal attacks by one or more parties. To be on fire.

Trolling (Wandering)

An inspiring (stimulating) post that is designed to provoke a flame war or flame responses. Trolling is basically sending messages that contain statements intending to temptation an argument or confrontation (fighting/war).

Messages that continue after the recipient asks that they stop are defined as a nuisance and are illegal. Nuisance messages that are threatening or intimidating are termed harassment and are even more illegal. It's interesting to note that nuisances and harassment are defined by the messages' *receiver*, not by the sender. That is, if you ask me to stop sending you messages, I have to even if I don't think there's anything wrong with my messages.

Acceptable Use Policy (AUP)

- ▶ outlines what types of uses are acceptable and which are prohibited although a particular institution's AUP will be unique, most have the following three categories (see next slides for more info) defining rights and responsibilities in using equipment in a legal and ethical manner legal documents that should be understood and followed

Personal Computer Files (documents)

- ▶ Storage location
- ▶ Storage space allocation
- ▶ Types of files allowed
- ▶ Printable items and quantity

Personal and Private Information

- ▶ Sharable information
- ▶ E-mail and personal file privacy
- ▶ Monitoring on-line activities

Ethical Conduct

- ▶ Care and consideration of equipment
- ▶ Respect rights of other users
- ▶ Policies concerning games, personal Web browsing, chat, and instant messaging
- ▶ Protect system security

- ▶ Software and plug-in installation
- ▶ Unacceptable topics, words, and graphics for papers, e-mail, and Web-page viewing (such as pornography, terrorism, illegal substances, etc.)

Code of Ethics

Many professional organizations have developed codes of ethics, which have been widely adopted as the “basic” code for many companies and universities. While the professional codes of ethics have slight differences in emphasis, they are in agreement on general principles.